**Shake It Off (The Budget Concerns)**:
Assessing the Economic Impact of Hosting Taylor Swift's Eras Tour

GPCO 454 - Quantitative Methods II - Homework 2

You've been working in the San Diego Mayor's Office, specifically in the Economic Planning Department, for a few months now. The work is meaningful, but some days feel long and routine. However, today is not one of those days. It started with a simple rumor, overheard while grabbing your morning coffee: Taylor Swift might be planning another concert tour within the next two years. As a die-hard Swiftie (though you'd never openly admit it at work), your mind immediately races. Could San Diego become one of the stops on her next tour?

You're already daydreaming about screaming along to All Too Well (10-Minute Version) when a thought hits you—this could be big for the city. Her recent Eras Tour set economic records, reportedly boosting local economies in each city she visited by millions of dollars.[1] The streets were packed with tourists, hotels sold out, and restaurants had wait times that stretched for hours. Could a Taylor Swift concert be just what San Diego's economy needs to drive growth and recovery after the challenges of recent years?

With renewed purpose, you pitch the idea to the mayor and the senior planning team at the next brainstorming session. You outline how the Eras Tour brought a surge in tourism, increased local business revenue, and generated tax dollars for other cities. The mayor listens but is skeptical. "Look," he says, "it's an intriguing idea, but you need to understand that hosting an event of this magnitude isn't cheap. We'd be talking about significant investments—venue security, public transportation upgrades, and permits, to name a few. And frankly, with urgent issues like wildfire recovery and homelessness on our plate, I can't justify allocating funds unless you can prove that hosting this concert will have a substantial economic impact."

**Challenge accepted.** You leave the meeting determined to make your case. To convince the mayor, you'll need more than anecdotes. You'll gather data on past concerts and apply multiple regression analysis to examine how major events impact local economies. Your task is to assess whether hosting concerts boosts the local economy. The fate of San Diego's potential Taylor Swift concert now rests in your hands. Can you prove it's more than just a fan's dream but an economic win for the city?

[1] See this, this, and this article, for example.

# 1 General Instructions

This homework is designed to help you apply concepts and tools to a real-world policy question. Your goal is to use data and multiple regression to analyze how hosting Taylor Swift's Eras Tour impacted the local (county-level) economy.

Here are a few key points to keep in mind:

1. **Focus on Clarity and Interpretation**: The primary objective is to interpret the results of your analysis in the context of the policy question. Explain your findings clearly and concisely.

2. **Follow Best Practices in R**: Use the coding techniques from lab sessions to manage data, run regressions, and produce clear outputs.

3. **Submission Format**: Submit two files electronically through Canvas:

- A PDF document (`LastName_PID_HW2.pdf`) containing your answers to all questions, including any written explanations, tables, and figures.
- A clearly labeled R script (`LastName_PID_HW2.R`) that contains all the code used to load data, run analyses, and generate figures and tables.

Both files must be well-organized, and your R script should run without errors. Assume the grader will download the script into the same directory as the data file. Your R script needs to run as a standalone file. Once complete, test your script by running it all the way through to ensure it works from start to finish without any pauses or interruptions.

4. **Collaboration Policy**: You may collaborate with others but must acknowledge all collaborators and tools. If you used AI tools like ChatGPT, specify how they were used.

**Example Acknowledgment (to be included at the beginning of your PDF submission):**

> **ACKNOWLEDGMENTS:** I, Alexander J. Norling, acknowledge working with John Doe and Jane Smith. I also used ChatGPT for help with data manipulation and improving plot aesthetics. All analyses and conclusions are my own.

5. **Late Submissions**: Late submissions will not be graded. Ensure that your files are submitted on time; the timestamp on the Canvas server will be used to determine timeliness.

6. **Ask for Help**: If you encounter any difficulties, don't hesitate to reach out during office hours or review sessions.

## 2 About the Data

For this assignment, you will work with three separate datasets:

- **County-level Annual GDP (2001-2023)**: Provided by the *Bureau of Economic Analysis (BEA)* Interactive Data Application. The dataset is contained in the file `HW2_CAGDP2_ALL_AREAS_2001_2023.xlsx`.
- **Taylor Swift's Eras Tour**: A dataset with information on the locations and dates of all U.S. tour stops, compiled by a dedicated Swiftie from an online forum. The dataset is contained in the file `HW2_The_Eras_Tour_US_Schedule.xlsx`.
- **County Demographics and Socioeconomic Indicators**: Provided by the *USDA Economic Research Service*, derived from the Small Area Income and Poverty Estimates (SAIPE) Program. Data includes the 2023 Rural-Urban Continuum Code, population estimates, net migration, education attainment levels, and poverty rates. The dataset is contained in the file `HW2_County_Demographics.xlsx`.

Details on variable names can be found in the accompanying codebooks, footnotes, and descriptions. These documents are provided in the following files:

- `HW2 Industry Category and Description.pdf`
- `HW2 CAGDP2 Footnotes.pdf`
- `HW2_CAGDP2_ALL_AREAS_2001_2023.xlsx`
- `HW2_County_Demographics.xlsx`
- `HW2_The_Eras_Tour_US_Schedule.xlsx`

Now is a good time to check all the files provided for this assignment. The Excel files contain multiple sheets, including codebooks and footnotes that explain key variables, definitions, and data formats. In practice, data in large projects is often presented in a similarly complex and detailed manner. While this may seem daunting at first, these documents are actually very helpful for ensuring your analysis is accurate. It's crucial to spend time understanding them to avoid errors. Open the datasets to get a sense of what's inside—variable names, data types, and any missing or special values. Remember, you cannot blindly proceed with your regression analysis without first comprehending the structure and content of your data.

## 3 Your Tasks

**1. R Script.** Your R script should thoroughly address **each and every numbered list** outlined in the tasks below. Document the steps you take in your script with meaningful comments that explain your code and provide context for each step. The goal is to ensure that someone with a reasonable understanding of R can follow your logic, understand how you implemented each task, and verify your results. Pay special attention to covering all aspects of the task instructions to demonstrate mastery of the concepts. Save your script as `LastName_PID_HW2.R` (e.g., `Ravanilla_A12345678_HW2.R`). Ensure there are no spaces in the filename.

**2. PDF Submission.** Your PDF submission should clearly and concisely answer all questions (e.g., **Q1**, **Q2**, etc.) listed under each section. Organize your answers logically and ensure that any figures or tables generated as part of your analysis are properly labeled, referenced, and explained. Save your PDF as `LastName_PID_HW2.pdf` (e.g., `Ravanilla_A12345678_HW2.pdf`). Ensure there are no spaces in the filename.

### 3.1 Preliminaries

1. Set your working directory and load all necessary packages in your R script. **IMPORTANT:** Comment out the `setwd()` command with a # before submitting your R script to save time when we run and check your script.

2. Load the following datasets into R:
   (1) `HW2_CAGDP2_ALL_AREAS_2001_2023.xlsx`,
   (2) `HW2_The_Eras_Tour_US_Schedule.xlsx`, and
   (3) `HW2_County_Demographics.xlsx`.
   Assign these datasets to variables called `cagdp_data`, `eras_data`, and `controls_data`, respectively.

3. Merge `controls_data` into `cagdp_data`. Name the resulting dataset `merged_data`.

4. Remove the variable `GeoName.y` from `merged_data` and rename `GeoName.x` to `GeoName`, since these are duplicate variables. Verify that the merge was successful by checking the structure of `merged_data`.

5. Filter `merged_data` to include only cases where `Rural_Urban_Continuum_Code_2023` equals 1. This will restrict the data to the most urban, metro areas with 1 million or more residents. Name this filtered dataset `filtered_data`.

6. Remove the variable `Rural_Urban_Continuum_Code_2023` from `filtered_data` to keep the dataset tidy.

7. Merge `eras_data` into `filtered_data`. Name the resulting dataset `working_data`.

8. Check the dimension of `working_data`.

9. Count the unique number of observations in the GeoName variable.

**Q1:** How many observations and variables are in the working dataset?

**Q2:** According to the U.S. Department of Agriculture's Economic Research Service, the Rural-Urban Continuum Codes (RUCC) classify U.S. counties into nine categories based on metropolitan status, population size, and degree of urbanization. Code 1 corresponds to counties in metropolitan areas with a population of 1 million or more. How many such counties are represented in the data?

**Q3:** If each row in the dataset represents a county-level observation, and there are only a few hundred counties in U.S. metropolitan areas with populations of 1 million or more, why does the `working_data` contain thousands of rows?

**Q4:** What do the variables `2001`, `2002`, up to `2023` represent in the working dataset at this point?

## 3.2 Data Cleaning: Dependent Variables

1. Identify all variables from `2001` to `2023` that contain county-level GDP data. You may want to store these variable names in a vector for easier processing.
   **Hint:** You can create this vector programmatically using
   `year_columns <- as.character(2001:2023)`. This converts the sequence of years from 2001 to 2023 into character strings that match the column names in your dataset. Having these stored in a vector will make it easier to apply transformations across multiple columns.

2. Replace all occurrences of the value `(D)` with `NA`.
   **Hint:** You can use `mutate()` along with `across()` to apply a function to multiple columns at once. For example, you may want to use:

   ```
   mutate(across(all_of(year_columns), ~ ifelse(.x == "(D)", NA, .x)))
   ```

3. Convert these variables to numeric data type after replacing `(D)` with `NA`.

## 3.3 Variable Creation: Hosting an Eras Tour Concert (Binary Indicator)

1. Create a new binary variable, `eras_tour_host`, that indicates whether a county hosted an Eras Tour concert in 2023. This variable should be equal to `1` if the county had at least one concert in 2023, and `0` otherwise.

## 3.4 Understanding the Outcome Variables

Large-scale events like the **Eras Tour** have the potential to impact local economies in direct and indirect ways. The *North American Industry Classification System (NAICS)* – represented by the variable `IndustryClassification` – helps categorize industries based on their primary activities, providing a detailed structure for economic analysis. In this section, we focus on which industries, based on their NAICS classifications, are most, moderately, and least likely to experience significant changes when a county hosts a major event. (Note: Sections 3.4.1 to 3.4.3 contain no tasks. They are provided for your reference only. Tasks begin in Section 3.4.4.)

### 3.4.1 Industries Most Likely to Be Impacted

- **Arts, Entertainment, and Recreation (NAICS 71)**:
  This includes concert venues, sports arenas, and entertainment services, which directly benefit from increased ticket sales and event participation.
- **Accommodation and Food Services (NAICS 72)**:
  Hotels, restaurants, and catering services often experience increased demand as attendees book accommodations and dine out during their visit.

### 3.4.2 Industries Moderately Likely to Be Impacted

- **Professional, Scientific, and Technical Services (NAICS 54)**:
  Event-related needs such as management, marketing, event planning, and technical support may increase temporarily but are less dependent on direct consumer activity.

### 3.4.3 Industries Least Likely to Be Impacted

- **Agriculture, Forestry, Fishing, and Hunting (NAICS 11)**:
  Events like the Eras Tour are unlikely to have any meaningful impact on agricultural production or services.

### 3.4.4 Transforming the Outcome Variables

1. Convert all GDP variables from `2001` to `2023` to per capita values. **NOTE:** Round the resulting values to two decimal places to improve readability.
2. Filter the dataset to include only rows where `IndustryClassification == 71`, narrowing the analysis to the *Arts, Entertainment, and Recreation* industry. This industry is expected to experience some of the most significant impacts from hosting large-scale events such as the Eras Tour.
3. Create a histogram to visualize the distribution of county-level GDP per capita for this industry in 2023. NOTE: Set binwidth = 0.5. Save it as an image file. item
4. Transform the GDP variables from `2001` to `2023`, which are already in per capita terms, into the log of (1 + thousand USD per capita) values.
5. Filter once again the dataset to include only rows for the Arts, Entertainment, and Recreation industry (NAICS 71), and then generate a histogram of the transformed variable, log of 2023 county-level GDP per capita in this industry. NOTE: Set binwidth = 0.5. Save it as an image file.

**Q5:** In this section, we applied two transformations to the outcome variables: first, we converted GDP to per capita by dividing it by population, and second, we transformed the per capita values by taking their logarithm. **Q5a:**Why was it important to adjust for population size when analyzing GDP? **Q5b:** Based on the histograms you generated, what issue in the data prompted the need for a log transformation? **Q5c:** Did the log transformation resolve the issue you observed in the initial histogram of GDP per capita? Briefly explain your reasoning.

## 3.5 Analyzing Time Trends in Arts, Entertainment, and Recreation Industry (NAICS 71) Annual Log GDP per Capita

In this section, you will create a plot to visualize the differences in economic activity (in log-transformed GDP per capita) between counties that hosted the Eras Tour and those that did not, across the years for which data is available.

1. **Filter the data:** Subset the dataset to include only rows where `IndustryClassification == 71`. This focuses your analysis on the *Arts, Entertainment, and Recreation* industry.

2. **Prepare the data for visualization:** Transform the data to a long format with years from 2001 to 2023 as a single variable. Then, calculate the **average log GDP per capita** for each year, grouped by the variable `eras_tour_host`.

3. Convert the `Year` variable to a numeric type to ensure proper ordering along the x-axis.

4. **Create the plot:** Generate a line plot with `ggplot2`. Use years (2001–2023) on the x-axis and the average log GDP per capita on the y-axis. Ensure the plot includes:

   - A title describing the trend comparison across both groups.
   - All years displayed on the x-axis without scientific notation.
   - A legend identifying each group (`eras_tour_host = 1` for counties that hosted the Eras Tour, and `eras_tour_host = 0` for those that did not).

5. **Save your plot:** Export the plot as an image file (e.g., PNG) to include in your final report. Use a descriptive file name such as `naics_71_log_gdp_plot.png`.

**Q6:** Examine the plot you generated. How does the trend in log GDP per capita for counties that hosted the Eras Tour compare to those that did not? Specifically, how do the slopes of these trends differ?

**Q7:** Why might this difference in pre-existing trends between the two groups create a problem when estimating the effect of hosting the Eras Tour on GDP?

## 3.6 Visualizing Demographic Differences Across Counties that Hosted the Eras Tour and Those That Did Not

In this section, you will create a series of bar plots to visualize key demographic and socioeconomic differences between counties that hosted the Eras Tour and those that did not. Specifically, you will analyze the following characteristics:

- Educational attainment shares:
  - Less than a high school diploma (2018–22)
  - High school diploma only (2018–22)
  - Some college or associate degree (2018–22)
  - Bachelor's degree or higher (2018–22)
- Poverty rate (2021)
- Net migration rate (2023)

To compare these characteristics, you will create bar plots. The treatment group (`eras_tour_host = 1`) will display 95% confidence intervals (CIs), while the control group (`eras_tour_host = 0`) will not. To simplify the process, the necessary R code to generate these plots is provided below. You can simply copy and run the code in your R script.

1. **Install and Load Necessary Packages**

   To proceed, you need the `patchwork` library to combine multiple plots into a panel layout. If you haven't installed it yet, use the following code:

```r
# Install patchwork if not already installed
install.packages("patchwork")

# Load required libraries
library(tidyverse)
library(patchwork)
```

2. **Prepare Population Characteristics for Visualization**

   Calculate the share of the population by education level, along with poverty and migration rates:

```r
# Define the share of the population by educational attainment
regression_data <- working_data %>%
  mutate(
    Less_than_HS_Share = (`Less than a high school diploma, 2018-22` / POP_ESTIMATE_2023) * 100,
    HS_Only_Share = (`High school diploma only, 2018-22` / POP_ESTIMATE_2023) * 100,
    Some_College_Share = (`Some college or associate's degree, 2018-22` / POP_ESTIMATE_2023) * 100,
    Bachelors_or_Higher_Share = (`Bachelor's degree or higher, 2018-22` / POP_ESTIMATE_2023) * 100
  )
```

3. **Summarize Data by Treatment Group**

Summarize the data by `eras_tour_host`, calculating 95% CIs only for the treatment group:

```
# Summarize data by eras_tour_host
summary_with_ci <- regression_data %>%
  group_by(eras_tour_host) %>%
  summarize(
    Less_than_HS = mean(Less_than_HS_Share, na.rm = TRUE),
    Less_than_HS_CI = ifelse(eras_tour_host == 1, 1.96 * sd(Less_than_HS_Share, na.rm = TRUE) / sqrt(n()), NA),

    HS_Only = mean(HS_Only_Share, na.rm = TRUE),
    HS_Only_CI = ifelse(eras_tour_host == 1, 1.96 * sd(HS_Only_Share, na.rm = TRUE) / sqrt(n()), NA),

    Some_College = mean(Some_College_Share, na.rm = TRUE),
    Some_College_CI = ifelse(eras_tour_host == 1, 1.96 * sd(Some_College_Share, na.rm = TRUE) / sqrt(n()), NA),

    Bachelors_Higher = mean(Bachelors_or_Higher_Share, na.rm = TRUE),
    Bachelors_Higher_CI = ifelse(eras_tour_host == 1, 1.96 * sd(Bachelors_or_Higher_Share, na.rm = TRUE) / sqrt(n()), NA),

    Poverty = mean(PCTPOVALL_2021, na.rm = TRUE),
    Poverty_CI = ifelse(eras_tour_host == 1, 1.96 * sd(PCTPOVALL_2021, na.rm = TRUE) / sqrt(n()), NA),

    Net_Migration = mean(NET_MIG_2023, na.rm = TRUE),
    Net_Migration_CI = ifelse(eras_tour_host == 1, 1.96 * sd(NET_MIG_2023, na.rm = TRUE) / sqrt(n()), NA),
    .groups = "drop"
  )
```

4. **Generate and Save the Bar Plots**

Use a function to create individual bar plots with 95% CIs for the treatment group:

```
# Function to create bar plots with 95% CI
plot_bar_with_ci <- function(variable, ci_variable, title, y_label) {
  plot_data <- summary_with_ci %>%
    filter(!is.na(.data[[variable]]))

  ggplot(plot_data, aes(x = as.factor(eras_tour_host), y = .data[[variable]], fill = as.factor(eras_tour_host))) +
    geom_bar(stat = "identity", position = "dodge", width = 0.7) +
    geom_errorbar(
      aes(ymin = .data[[variable]] - .data[[ci_variable]],
          ymax = .data[[variable]] + .data[[ci_variable]]),
      width = 0.2,
      data = filter(plot_data, eras_tour_host == 1)
    ) +
    labs(title = title, x = "Eras Tour Host", y = y_label,
      fill = "Eras Tour Host") +
    theme_minimal() +
    theme(plot.title = element_text(size = 10), axis.text.x =
        element_text(angle = 45, hjust = 1))
}

# Generate plots for each characteristic
plot1 <- plot_bar_with_ci("Less_than_HS", "Less_than_HS_CI", "Less than High School Diploma", "Percentage (%)")
plot2 <- plot_bar_with_ci("HS_Only", "HS_Only_CI", "High School Diploma Only", "Percentage (%)")
plot3 <- plot_bar_with_ci("Some_College", "Some_College_CI", "Some College or Associate Degree", "Percentage (%)")
plot4 <- plot_bar_with_ci("Bachelors_Higher", "Bachelors_Higher_CI", "Bachelor's Degree or Higher", "Percentage (%)")
plot5 <- plot_bar_with_ci("Poverty", "Poverty_CI", "Poverty Rate (2021)", "Percentage (%)")
plot6 <- plot_bar_with_ci("Net_Migration", "Net_Migration_CI", "Net Migration (2023)", "Net Migration Rate")

# Arrange all plots in a 3x2 panel and display
```

```r
final_panel <- (plot1 | plot2) / (plot3 | plot4) / (plot5 | plot6)
print(final_panel)

# Save the final plot as an image file
ggsave("demographics_comparison_panel.png", plot = final_panel, width = 12, height = 10, dpi = 300)
```

**Q8:** Examine the bar plots you generated. How do the demographic and socioeconomic characteristics of counties that hosted the Eras Tour compare to those that did not? Specifically, which characteristics show notable differences between the two groups?

**Q9:** Why might these differences in demographic and socioeconomic characteristics between the two groups create a problem when estimating the effect of hosting the Eras Tour on GDP?

## 3.7 Multiple Regression Analyses

### 3.7.1 Multiple Regression Analyses: Industries Most Likely to Be Impacted (NAICS 71)

In this section, you will perform a series of multiple regression analyses to examine the impact of hosting the Eras Tour on log GDP per capita in the *Arts, Entertainment, and Recreation* industry (NAICS 71). The goal is to progressively add control variables to understand how these factors affect the estimated relationship between hosting the Eras Tour and economic outcomes.

1. **Filter the data:** Subset the dataset to include only rows where the `IndustryClassification` corresponds to NAICS 71.

2. **Prepare relevant variables:** Create variables representing the share of educational attainment levels, poverty rate, and net migration rate. These will serve as control variables in your regression models.

   **Hint:** Use `mutate()` to calculate the percentage shares and select only relevant variables for your analysis.

3. **Define regression models:** You will define five models with progressively added controls. Start with a simple regression of log GDP per capita for 2023 on the treatment variable (`eras_tour_host`). Add educational attainment (various shares), poverty, net migration, and GDP for 2022 as controls in subsequent models.

4. **Generate the regression table:** Summarize the results of all five models in a single table. The dependent variable for all models is log GDP per capita in 2023.

### 3.7.2 Multiple Regression Analyses: Industries Most Likely to Be Impacted (NAICS 72)

In this section, you will perform multiple regression analyses for the *Accommodation and Food Services* industry (NAICS 72). Repeat the process you followed in the previous section, but this time, filter the dataset to include only rows where the `IndustryClassification` corresponds to NAICS 72.

1. **Filter the data:** Subset the dataset to include only rows for NAICS 72.

2. **Prepare relevant variables:** Calculate the control variables as you did for NAICS 71, including educational attainment shares, poverty rate, and net migration rate.

3. **Define regression models:** Run a series of regression models with progressively added controls. Use log GDP per capita for 2023 as the dependent variable.

4. **Generate the regression table:** Summarize the results of all models in a table similar to the one you generated for NAICS 71.

**Hint:** You can reuse most of the R code from the previous section by adjusting the `IndustryClassification` filter to NAICS 72.

### 3.7.3 Multiple Regression Analyses: Industries Moderately Likely to Be Impacted (NAICS 54)

In this section, you will conduct multiple regression analyses for the *Professional, Scientific, and Technical Services* industry (NAICS 54). Repeat the steps from the previous section, adjusting the dataset to include only rows where the `IndustryClassification` corresponds to NAICS 54.

1. **Filter the data:** Subset the dataset to include only rows for NAICS 54.
2. **Prepare relevant variables:** Generate the control variables such as educational attainment shares, poverty rate, and net migration rate.
3. **Define regression models:** Perform a series of regressions using log GDP per capita for 2023 as the dependent variable. Add controls progressively across models.
4. **Generate the regression table:** Summarize the results for all models in a table, similar to the one produced in earlier sections.

**Hint:** You can use the same code structure from the previous sections, making sure to update the filter to `IndustryClassification == 54`.

### 3.7.4 Multiple Regression Analyses: Industries Least Likely to Be Impacted (NAICS 11)

In this section, you will conduct multiple regression analyses for the *Agriculture, Forestry, Fishing, and Hunting* industry (NAICS 11). Follow the process outlined in the previous sections, filtering the dataset for NAICS 11.

1. **Filter the data:** Subset the dataset to include only rows for NAICS 11.
2. **Prepare relevant variables:** Generate control variables such as educational attainment shares, poverty rate, and net migration rate.
3. **Define regression models:** Perform a series of regressions with log GDP per capita for 2023 as the dependent variable. Progressively add controls in each model.
4. **Generate the regression table:** Summarize the results for all models in a table, following the format used in earlier sections.

**Q10:** Look at the coefficient for `Eras Tour Host` in Model 5 under NAICS 71. What does the estimate tell you about how hosting the Eras Tour affects county-level GDP per capita in the Arts, Entertainment, and Recreation industry? Is the coefficient statistically significant?

**Q11:** How does controlling for GDP per capita in 2022 in Model 5 help address the issue raised in **Q7** regarding pre-existing trends? Why is this step important in reducing omitted variable bias?

**Q12:** How do the changes in the coefficient estimates for `eras_tour_host` across the models address the issue raised in Q9 regarding omitted variable bias from demographic and socioeconomic differences?

**Q13:** Examine the regression results for the other industries (NAICS 72, NAICS 54, and NAICS 11). Are the results consistent with the hypothesis that the Eras Tour had a greater economic impact on industries most likely to benefit from large events? Provide evidence from the models to support your response.

**Q14:** Based on your regression analysis for the Arts, Entertainment, and Recreation industry (NAICS 71), calculate the implied increase in GDP for counties that hosted the Eras Tour. Use the provided hints and data to guide your calculations.

**Hint 1:** Refer to the coefficient for `Eras Tour Host` in Model 5 under NAICS 71. Since the outcome is in log GDP per capita, follow these steps:

1. **Exponentiate the coefficient**: Convert the log effect to a proportional change.
2. **Find the percentage increase**: Subtract 1 from the exponentiated value and multiply by 100.
3. **Calculate the estimated increase in thousand USD**: Multiply the percentage increase by the average NAICS 71 GDP in host counties, which is 4,100,000 thousand USD.
4. **Calculate the per capita dollar impact**: Use the average county population of 2,300,000 to estimate the economic boost per capita.

**Q15:** Now, based on your calculation in **Q14**, what would you report to the mayor regarding the potential economic impact of hosting a Taylor Swift concert in San Diego? Provide a structured response that addresses the following points:

1. **Evidence of economic impact**: Summarize the key findings from the regression analysis, including the size and statistical significance of the effect.
2. **Estimated economic benefit**: Report the dollar increase in county-level GDP per capita and total GDP implied by the model.

3. **Strength of the argument**: Explain how controlling for education, poverty, migration, and prior GDP helps provide a more credible estimate of the impact, reducing concerns about omitted variable bias.
4. **Relevance to San Diego**: Discuss how these findings support the idea that a major event could boost the local economy and provide a rationale for investing in hosting the concert mindful of the cost.